FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA TEST SERIES

FIITJEE JEE (Advanced), 2015

CONCEPT RECAPITULATION TEST - IV

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 180**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 10)** contains 10 multiple choice questions which have one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (11 to 16) contains 3 paragraphs with each having 2 questions. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (17 – 20) contains 4 Matching Lists Type questions: Each question has four statements in LIST I & 4 statements in LIST II. The codes for lists have choices (A), (B), (C), (D) out of which only one is correct. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

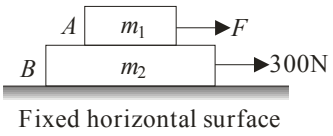
SECTION – A

(Only One Option Correct Type)

This section contains **10 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** option **is correct**.

1. Electrons in a sample of gas containing hydrogen like atom (Z = 3) are in fourth excited state. When photons emitted only due to transition from third excited state to second excited state are incident on a metal plate photoelectrons are ejected. The stopping potential for these photoelectrons is 3.95 eV. Now, if only photons emitted due to transition from fourth excited state to third excited state are incident on the same metal plate, the stopping potential for the emitted photoelectrons will be approximately equal to:
(A) 0.85 eV (B) 0.75 eV
(C) 0.65 eV (D) None of these

2. The coefficient of friction between block *A* of mass m_1 = 5 kg and block *B* of mass m_2 = 10 kg is μ = 0.5. There is no friction force between block *B* and fixed horizontal surface. A force of 300 N acts on block *B* in horizontal direction and a horizontal force of magnitude *F* acts on block *A* as shown, both towards right. Initially there is no relative motion between the blocks. The minimum value of *F* such that relative motion starts between *A* and *B* is:

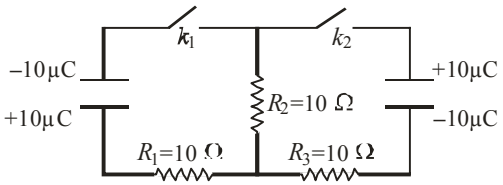


(A) 200N (B) 187.5N
(C) 150 N (D) 0

3. A proton moves in the positive *z*-direction after being accelerated from rest through a potential difference *V*. The proton then passes through a region with a uniform electric field *E* in the positive *x*-direction and a uniform magnetic field *B* in the positive *y*-direction, but the proton's trajectory is not affected. If the experiment were repeated using a potential difference of 2V, the proton would then be:
(A) deflected in positive *x*-direction (B) deflected in negative *x*-direction
(C) deflected in positive *y*-direction (D) deflected in negative *y*-direction

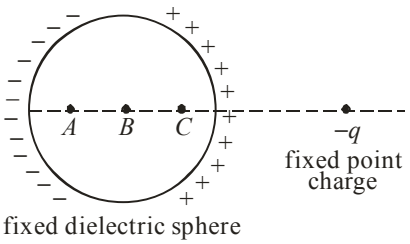
Space for Rough work

4. In the circuit shown initially both keys are open and capacitance of both capacitors is 1 μF. Charges on each capacitor are as shown. Now both the keys k_1 and k_2 are simultaneously closed. Just after closure of both keys, the current through resistance R_2 is:



(A) 1 ampere
(B) 0.5 ampere
(C) 1.5 ampere
(D) zero

5. A negative point charge is brought near an uncharged solid dielectric sphere. Positive induced charges appear on the side of the sphere nearest to the point charge, negative induced charges on the opposite side. *A*, *B* and *C* are three points on diameter of the sphere that also passes through the point charge. Comparing only at points *A*, *B* and *C* on the dielectric sphere, the potential is:



(A) Largest at *C*
(B) Largest at *A*
(C) Largest at *B*
(D) The same every where

6. A uniform thin, rod *AB* of length *L* and mass *m* is undergoing fixed axis rotation about end *A*, such that end *A* remains stationary as shown. The kinetic energy of section *AP* of rod is equal to kinetic energy of section *BP* of rod at an instant. Then the ratio of length *AP* and length *AB*, that is, $\frac{AP}{AB}$:

(A) $\frac{1}{2}$
(B) $\frac{1}{2^{1/3}}$
(C) $\frac{1}{\sqrt{2}}$
(D) $\frac{1}{2\sqrt{2}}$

Space for Rough work

7. In displacement method, the distance between object and screen is 96 cm. The ratio of length of two images formed by a convex lens placed between them is 4:
(A) Ratio of the length of object to the length of shorter image is 2.
(B) Distance between the two positions of the lens is 40 cm.
(C) Focal length of the lens is 64 cm.
(D) When the shorter image is formed, distance of the lens from the shorter image is 64 cm.

8. An unstable radioactive element *A* decays into a unstable daughter element *B* which in turn decays into stable grand-daughter element *C*. At time $t = 0$, the element *A* consists of N_0 atoms while number of atoms of element *B* and element *C* are zero. The decay constant λ of *A* and *B* are same. Based on this situation choose the incorrect statement from the following:
(A) At the instant activity of element *B* is zero, the number of atoms of element *B* are maximum.
(B) The maximum number of atoms of element *B* present at any moment is less than $\frac{N_0}{2}$.
(C) At the instant activity of element *B* is maximum, the activity of element *A* and element *B* are equal.
(D) The activity of stable element *C* increases with time.

9. Which of the following statements is/are incorrect:
(A) A positively charged particle having non-zero initial velocity may move from region of lower electric potential towards region of higher electric potential.
(B) Whenever a charged solid conductor is grounded, the net charge on conductor must become zero.
(C) Even though the plates of a parallel plate capacitor are oppositely charged, the net electrostatic potential energy stored inside the capacitor is positive.
(D) As the inter-plate distance of a parallel plate capacitor is slowly reduced to half its initial value, the electric field inside the capacitor may remain same.

Space for Rough work

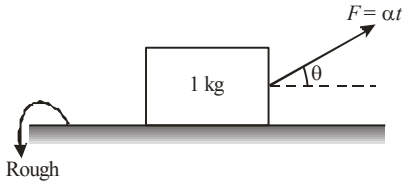
10. Which of the following statements is/are correct:
(A) If distance x between two particles at any time t is increasing then the distance between particles will be least if and only if $\frac{dx}{dt} = 0$.
(B) For a particle moving with constant acceleration, the rate of change of speed with time must be a constant.
(C) The magnitude of average velocity of a particle moving along a straight line between time $t = 0$ to $t = T$ may be greater than speed of particle at time $t = T$.
(D) Several stone are projected from top of a tower with same speeds but at different angles of projection. The velocities of all these stone just before reaching the ground are equal.

Comprehension type (Only One Option Correct)

This section contains **3 paragraphs**, each describing theory, experiments, data etc. Six questions relate to the three paragraphs with **two questions on each paragraph**. Each question has **only one** correct answer among the four given options (A), (B), (C) and (D).

Paragraph for Questions 11 & 12

A variable force $F = \alpha t$ is acting on block of mass 'm' at an angle θ with horizontal as shown.



(given m = 1 kg, α = 3 N/sec, $\mu_S = 0.75$, $\mu_K = \frac{1}{3}$, sin 53 = $\frac{4}{5}$). Give answer of following.

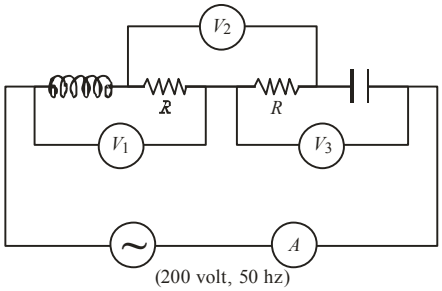
11. Minimum time after which block starts moving:
(A) 1 sec
(B) 2 sec
(C) 3 sec
(D) Data insufficient as α is not mentioned

12. Speed of block when it leaves the horizontal surface for $\theta = 37^\circ$ approximately:
(A) 20 m/s (B) 25 m/s
(C) 28 m/s (D) 26 m/s

Space for Rough work

Paragraph for Questions 13 & 14

In the shown *AC, L-C-R* series circuit reading of ideal voltmeter V_1 and V_3 are 300 volt each and reading of hot wire ammeter *A* is 10 amp then give the answer of followings.

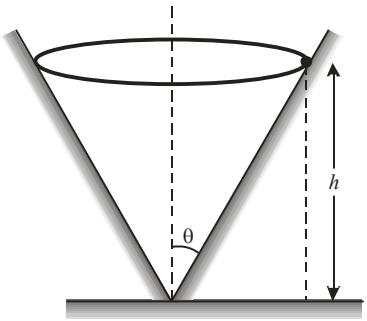


13. Reading of V_2 will be:
(A) 600 volt (B) 300 volt
(C) 400 volt (D) 200 volt

14. Value of resistance '*R*' will be:
(A) 10 Ω (B) 20 Ω
(C) 30 Ω (D) data insufficient

Paragraph for Questions 15 & 16

A particle of mass '*m*' is moving in horizontal circle inside a smooth inverted vertical cone above height '*h*' from apex. Half angle of cone is 'θ' then give the answer of following.



15. Normal force by surface of cone is :
(A) mg cos θ (B) mg sec θ
(C) mg sin θ (D) mg cosec θ

16. Time period of one revolution if θ increases keeping '*h*' constant :
(A) will remain same (B) will decrease
(C) will increase (D) increase then decrease

Space for Rough work

(Matching List Type)

This section contains **4 multiple choice questions. Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

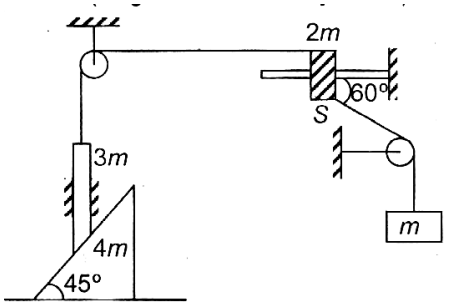
17. Match the statements from List I with those in List II and select the correct answer using the code given below the lists.

	List-I		**List-II**
(P)	Floating of a block in liquid	(1)	Work energy theorem
(Q)	The expression for buoyant force	(2)	Mass conservation
(R)	Equation of continuity	(3)	Equilibrium of a body
(S)	Bernoulli's theorem	(4)	Archimedes' principle

	P	Q	R	S
(A)	1	2	3	4
(B)	2	1	4	3
(C)	4	1	3	2
(D)	3	4	2	1

Space for Rough work

18. Consider an arrangement shown in the figure. The rod of mass 3m is constrained to move in the vertical direction only. The wedge of 4m moves in the horizontal direction. The slider S of mass 2m moves on a fixed horizontal rod. At a particular instant system is released and the string connecting the slider S to the making angle 60^0 with the rod. (Neglect friction everywhere) Match the statements from List I with those in List II and select the correct answer using the code given below the lists.

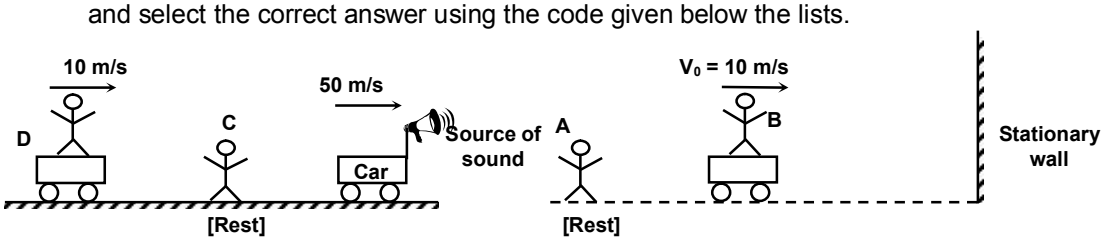


List– I			List - II
(P)	If net force on the block of m at this instant is $\frac{5amg}{37}$, then a =	(1)	3
(Q)	If net force on the wedge is $\frac{5bmg}{37}$, then b =	(2)	1
(R)	If contact force between wedge and rod is $\frac{5\sqrt{2}cmg}{37}$, then c =	(3)	2
(S)	If tension in the string between blocks having mass m and 2m is $\frac{21dmg}{37}$, then d =	(4)	8

	P	Q	R	S
(A)	1	2	3	4
(B)	2	4	4	3
(C)	4	1	3	2
(D)	3	4	1	2

Space for Rough work

19. A source of sound, emitting frequency of 3000 Hz, moving towards a stationary reflecting wall with speed 50 m/s. There are four observers A, B, C and D as shown in the figure. (Detectors A & C are at rest and detectors B & D are moving with speed 10 m/s towards right. Speed of sound is 350 m/s) Match the statements from List I with those in List II and select the correct answer using the code given below the lists.



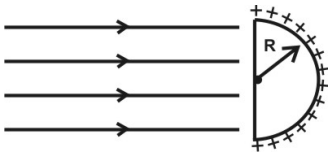
List-I contains detectors and List-II contains the beat frequency observed by the detectors. Match the List-I with List-II.

List-I		List-II	
(P)	Detector A	(1)	200 Hz
(Q)	Detector B	(2)	Zero
(R)	Detector C	(3)	900 Hz
(S)	Detector D	(4)	875 Hz

	P	Q	R	S
(A)	1	2	3	4
(B)	2	1	4	3
(C)	4	1	3	2
(D)	3	4	1	2

Space for Rough work

20. A semi circular disc of mass M and radius R with linear charge density λ on its curved circumference is hinged at its centre and placed in a uniform electric field as shown in the figure



Match the statements from List I with those in List II and select the correct answer using the code given below the lists.

List – I		List – II	
(P)	The net force acting on the ring is	(1)	$\pi\sqrt{\frac{M}{\lambda E}}$
(Q)	If the ring is slightly rotated about O and released, find its time period of oscillation	(2)	$\pi R\lambda E$
(R)	The work done by an external agency to rotate it through an angle α is	(3)	$E-\frac{2K\lambda}{R}$
(S)	Magnitude of electric field at 'O' will be	(4)	$4\lambda ER^2\sin^2\left(\frac{\alpha}{2}\right)$

	P	Q	R	S
(A)	1	2	3	4
(B)	2	4	1	3
(C)	2	1	4	3
(D)	3	4	1	2

Space for Rough work

Chemistry

PART – II

SECTION – A

(Only One Option Correct Type)

This section contains **10 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** option **is correct**.

1. The solution of weak acid $H_2N_2O_2$ decomposes spontaneously at room temperature. The gaseous product obtained is
   (A) NO (B) N_2O
   (C) N_2 (D) NO_2

2. The correct statement for different solvents at constant temperature is
   (A) The higher is the molar mass of solvent larger is K_b.
   (B) The higher is the molar mass of solvent smaller is K_b.
   (C) The higher is the molar mass of solvent larger is K_f.
   (D) The higher is the molar mass of solvent smaller is K_f.

3. The incorrect statement amongst the following is?
   (A) Black phosphorous is thermodynamically most stable allotrope of phosphorous
   (B) Fe(III) is thermodynamically more stable than Fe(II)
   (C) Graphite is thermodynamically more stable than diamond
   (D) White phosphorous is kinetically least stable allotrope of phosphorous and so is graphite in comparison to diamond
   (Note: All comparisons are in their respective standard states)

4. A certain amount of $N_2O_4(g)$ is enclosed in a closed container at 127°C when following equilibrium got set up at a total pressure of 10 atm.
   $N_2O_4(g) \rightleftharpoons 2NO_2(g)$
   If the concentration of $NO_2(g)$ in the equilibrium mixture be 8×10^5 ppm, the K_c (in mol L^{-1}) for the above reaction at 127°C is equal to
   (A) 3.189 (B) 2.051
   (C) 0.974 (D) 1.842

5. A metal M and its compounds can give the following observable changes in sequence of reactions.
   $$M \xrightarrow[HNO_3]{dil} \begin{bmatrix}\text{colourless}\\ \text{solution}\end{bmatrix} \xrightarrow[NaOH]{dil} \begin{bmatrix}\text{white}\\ \text{ppt}\end{bmatrix} \xrightarrow[NaOH]{excess} \begin{bmatrix}\text{colourless}\\ \text{solution}\end{bmatrix} \xrightarrow{H_2S} \begin{bmatrix}\text{white}\\ \text{ppt}\end{bmatrix}$$
   The metal M is
   (A) Mg (B) Pb
   (C) Zn (D) Sn

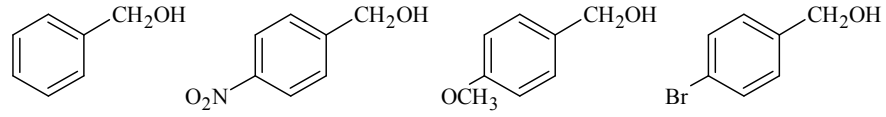
Space for Rough work

6. What is not correct about E_1CB reaction?
(A) Mechanism is followed when substrate is having excessive acidic proton while leaving group is a bad leaving group
(B) the rds is unimolecular and the intermediate is carbanionic
(C) the overall order of reaction is one
(D) the rate law of the reaction is as follows:

Rate $\propto \frac{[S]^2}{[B^-]}$, where S is the substrate and B^- is the base

7. $Fe_2O_{3(s)} + 3CO(g, 1\ atm) \longrightarrow 2Fe_{(s)} + 3CO_2\ (g, 1\ atm)$, $\Delta G_{25^\circ} = -10$ kJ
The ΔG for the reaction
$2Fe_{(s)} + 3CO_2\ (g, 2\ atm) \longrightarrow Fe_2O_{3(s)} + 3CO\ (g, 20\ atm)$
at the temperature 25°C is equal to
(A) +17.11 kJ (B) –7.11 kJ
(C) +27.11 kJ (D) – 5.70 kJ

8. Consider the following alcohols:



I II III IV

The order of decreasing reactivities of these alcohols towards substitution with HBr is:
(A) III > I > IV > II (B) III > I > II > IV
(C) I > III > IV > II (D) I > III > II > IV

9. Which of the following is a pyrosilicate.
(A) $Sc_2Si_2O_7$ (B) Zn_2SiO_4
(C) $Ca_2Si_3O_9$ (D) $Be_3Al_2Si_6O_{18}$

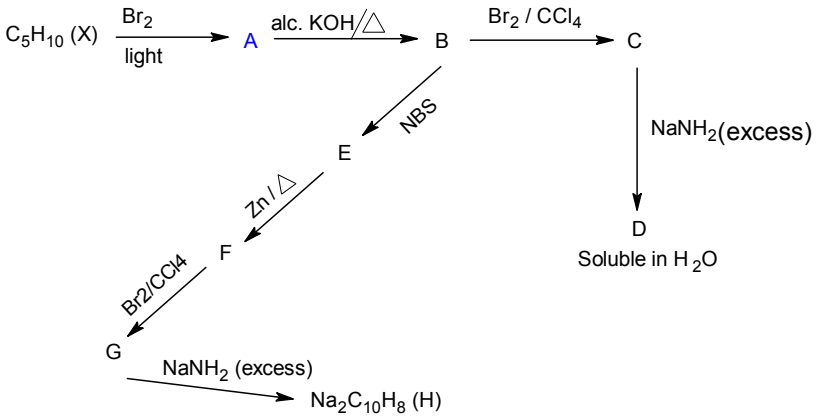
10. Boric acid is a weak acid but it behaves as a strong acid in presence of
(A) Glycerol (B) Borazole
(C) $FeCl_3$ (D) $AlCl_3$

Space for Rough work

Comprehension Type

This section contains **3 paragraphs** each describing theory, experiment, data etc. **Six questions** relate to four paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct** answer among the four choices (A), (B), (C) and (D).

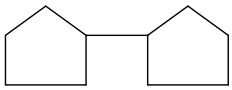
Paragraph for Question Nos. 11 to 12

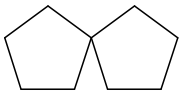


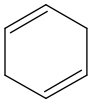
X is a saturated hydrocarbon which gives only one monobromination product:

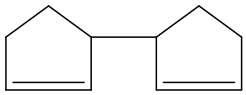
11. What is correct about D?
(A) D is an alkadiene
(B) D is an alkyne
(C) D is an aromatic ionic compound
(D) D can exist as two stereo isomeric forms

12. 'F' is:

(A) 

(B) 

(C) 

(D) 

Space for Rough work

Paragraph for Question Nos. 13 to 14

In coordination chemistry there are a variety of methods applied to find out the structures of complexes. One method involves treating the complex with known reagent and from the nature of reaction, the formula of the complex can be predicted. An isomer of the complex $Co(en)_2(H_2O)Cl_2Br$, on reaction with concentrated H_2SO_4 (dehydrating agent) suffers loss in weight and on reaction with $AgNO_3$ solution it gives a white precipitate which is soluble in $NH_3(aq)$.

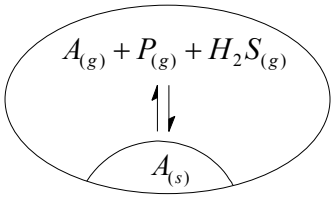
13. The correct formula of the complex is
(A) $[CoClBr(en)_2]H_2O$ (B) $[CoCl(en)_2(H_2O)]BrCl$
(C) $[CoBr(en)_2(H_2O)]Cl_2$ (D) $[CoBrCl(en)_2]Cl.H_2O$

14. If one mole of original complex is treated with excess $Pb(NO_3)_2$ solution, then the number of moles of white precipitate formed under cold conditions will be
(A) 0.5 (B) 1.0
(C) 0.0 (D) 3.0

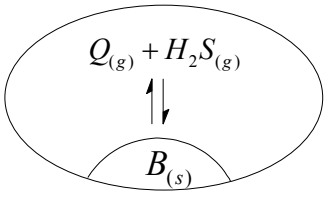
Space for Rough work

Paragraph for Question Nos. 15 to 16

Two identical bulbs contain an excess of solid A (volatile) and solid B (non-volatile) respectively.



I



II

At 25°C equilibrium gets established as shown above. At equilibrium; total pressure inside both bulbs is 10 mm whereas partial pressure of H_2S inside bulb I & II is 4 and 5 mm of Hg respectively.

Equilibria shown for bulb –I is a mix of

$A_{(s)} \xrightleftharpoons{\text{Decomosition}} P_{(g)} + H_2S_{(g)}$

$A_{(s)} \xrightleftharpoons{\text{Sublimation}} A_{(g)}$

Equilibrium shown for bulb –II

$B_{(s)} \xrightleftharpoons{\text{Decomosition}} Q_{(g)} + H_2S_{(g)}$

15. If some A(g) is removed from bulb–I at equilibrium:
(A) decomposition equilibrium will shift forward
(B) sublimation equilibrium will shift forward
(C) both decomposition and sublimation equilibrium will shift forward
(D) no change will happen in the state of equilibrium

16. If a mixture of excess solids A & B is kept at 25°C in a sealed container, the pressure developed inside the container at equilibrium will be:
(A) 2.5 mm of Hg (B) 3.9 mm of Hg
(C) 14.8 mm of Hg (D) 8.4 mm of Hg

Space for Rough work

(Matching List Type)

This section contains **4 multiple choice questions. Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

17. Match the following:-

	List-I (Binary liquid mixture in ether)		**List -II** (method used for separation)
(P)	Aniline + N-methyl aniline	(1)	(i) addition of $[Ag(NH_3)_2]^+$ (ii) filtration (iii) treating precipitate with dil HCl
(Q)	Phthalimide + benzyl amine	(2)	(i) addition of $C_6H_5SO_2Cl$ & aq NaOH (ii) Treating aqueous layer with dil HCl, con. HCl and Distilled over NaOH
(R)	1-propanol **+** propanone	(3)	(i) addition of NaOH (ii) treating aqueous layer with HCl
(S)	1-Butyne + 2-butyne	(4)	(i) addition of saturated $NaHSO_3$ (ii) filtration (iii) heating precipitate with dil HCl

	P	**Q**	**R**	**S**
(A)	3	2	4	1
(B)	4	3	2	1
(C)	2	3	4	1
(D)	3	4	2	1

18. Match the following:

	List – I		**List – II** (Nature of oxide product)
(P)	$NH_4NO_3 \xrightarrow[\Delta]{\text{gently}}$	(1)	Acidic & diamagnetic
(Q)	$NH_3 + O_2 \xrightarrow[\Delta]{Pt}$	(2)	Acidic & paramagnetic
(R)	$Cu + \underset{(Conc)}{HNO_3} \rightarrow$	(3)	Neutral & paramagnetic
(S)	$HNO_3 \xrightarrow[\Delta]{P_4O_{10}}$	(4)	Neutral & diamagnetic

	P	**Q**	**R**	**S**
(A)	1	2	3	4
(B)	3	4	1	2
(C)	4	2	1	3
(D)	4	3	2	1

Space for Rough work

19. Match the column.

List – I		List – II	
(P)	$CH_3COCH_2COOC_3H_7$	(1)	Gives positive iodoform test.
(Q)	$CH_3CH_2CH_2CHO$	(2)	Gives 2, 4–DNP derivative.
(R)	$CH_3CH(OH)CH=CH_2$	(3)	Decolourises Baeyer's reagent.
(S)	$C_2H_5OC(CH_3)=CH_2$	(4)	Do not give positive iodoform test.
		(5)	Can not show tautomerism.

	P	Q	R	S
(A)	2, 4	2, 4	3, 5	4, 5
(B)	2, 4	4	1,5	3, 4
(C)	2, 4	2, 4	1, 3, 5	3, 4, 5
(D)	2, 4	2	1, 3, 5	4, 5

20. Match the following:

	List – I		List - II
(P)	Plot of log $t_{1/2}$ vs log $[A_0]$ is a straight line with a slope of '-1'	(1)	Zero order
(Q)	Plot of log $t_{1/2}$ vs log $[A_0]$ is a straight line with a slope of '1'	(2)	2nd order
(R)	Plot of log K_p vs 1/T is a straight line with a negative slope	(3)	ΔH^0 is negative
(S)	Plot of log K_p vs 1/T is a straight line with a positive slope	(4)	ΔH^0 is positive

	P	Q	R	S
(A)	1	2	3	4
(B)	2	1	3	4
(C)	1	2	4	3
(D)	2	1	4	3

Space for Rough work

Mathematics PART – III

SECTION – A

(Only One Option Correct Type)

This section contains **10 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** option **is correct**.

1. If $f(x)=\begin{cases} -x^2, & x \geq 0 \\ (a^2-2a)+(b^2-2)x, & x<0 \end{cases}$ and $x=0$ is point of maxima, then

(A) $a \in (0, 2)$ for $b \in (-\sqrt{2}, \sqrt{2})$ (B) $a \in [0, 2]$ for $b \in R-[-\sqrt{2}, \sqrt{2}]$

(C) $a \in \{0, 2\}$ for $b \in R-(\sqrt{2}, \sqrt{2})$ (D) $a \in R-(0, 2)$ for $b \in R-[-\sqrt{2}, \sqrt{2}]$

2. If $P_n = 1+\cos 2^{n-1}\theta - \sin 2^{n-1}\theta$ and $Q_n = 1+\tan 2^{n-1}\theta$ such that $(P_1Q_0)\times(P_2Q_1)\times.....(P_nQ_{n-1})=1$, then θ may be

(A) $\frac{\pi}{64}$ (B) $\frac{\pi}{256}$

(C) $\frac{\pi}{33}$ (D) $\frac{\pi}{31}$

3. If $h(x)=f(x)+g(1-x)$ is an increasing function from $[0, \infty)$ to $[0, \infty)$ such that $g'(1-x)=f''(x)$ and $f'(0)=0$, then

(A) $g'(1-x)\leq 0$ for $x \geq 0$ (B) $g'(1-x)\geq 0$ for $x \geq 0$

(C) $g'(1-x)\leq 0$ for $x \leq 0$ (D) $g'(1-x)\geq 0$ for $x \leq 0$

4. If $f(x)=x+1$, $g^{-1}(x)=x^3+x+1$ and $g(f(x))=h(x)$, then $\lim\limits_{x\to 1}\frac{h^{-1}(x)-h^{-1}(1)}{x-1}$ is equal to

(A) 2 (B) 4

(C) 1 (D) 8

5. If $f(x)=\tan x+\tan^2 x\tan 2x$ and $g(n)=\sum\limits_{r=0}^{r=n} f(2^r)$, then $g(2013)-\tan(2^{2014})$ is equal to

(A) 0 (B) tan1

(C) $-\tan 1$ (D) 1

Space for rough work

6. If XY-plane is rotated about origin by an angle α in a.c.w. manner $\left(0 < \alpha < \frac{\pi}{2}\right)$, then the equation of line $\frac{x}{a} + \frac{y}{b} = 1$ is changed by $x\cos\alpha + y\sin\alpha = p$. Then, correct statement is

(A) $a = p\cos\alpha$, $b = p\sin\alpha$ (B) $a = p\sec\alpha$, $b = p\operatorname{cosec}\alpha$

(C) $\sqrt{\frac{1}{p^2} + \frac{2}{ab}} = \left|\frac{1}{a} + \frac{1}{b}\right|$ (D) $\frac{2}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}$

7. If $y = \cfrac{x}{x^2 + \cfrac{x}{x^2 + \cfrac{x}{x^2 + ...\infty}}}$ and $\int \frac{\left(y - x^2\right)dx}{\left(x^2 + y\right)\left(x + y^2\right)} = f\left(y\right) + c$, then f′(y) for $x = 1$ is equal to

(A) 1 (B) $\frac{\sqrt{5} - 1}{2}$

(C) $\frac{\sqrt{5} + 1}{2}$ (D) $\frac{2 - \sqrt{5}}{2}$

8. Suppose f and g are functions having second derivatives f" and g" everywhere, if f(x) g(x) = 1 for all x and f ' (x) and g' are never zero, then $\frac{f''(x)}{f'(x)} - \frac{g''(x)}{g'(x)}$ equals

(A) $\frac{f'(x)}{f(x)}$ (B) $2\frac{f'(x)}{f(x)}$

(C) $-\frac{2g'(x)}{g(x)}$ (D) $-\frac{2f'(x)}{f(x)}$

9. Let A be the point of intersection of the curve $y = \log_{\frac{1}{3}}\left(x - \frac{1}{3}\right) + \log_3\sqrt{9x^2 - 6x + 1}$ and the circle $x^2 + y^2 = 17$. B is also a point on the curve but lies inside the given circle such that its abscissa is an integer, then max {AB} is

(A) 2 units (B) 3 units

(C) 4 units (D) none of these

10. If S_n denotes the sum of first n terms of an AP and $a_n = \frac{S_{3n}}{S_{2n} - S_n}$, then $\underset{n\to\infty}{\text{Lt}} \sum_{r=1}^{2n} \frac{a_r}{n}$ is equal to

(A) 2 (B) 4

(C) 6 (D) 8

Space for rough work

Comprehension type (Only One Option Correct)

This section contains **3 paragraphs**, each describing theory, experiments, data etc. Six questions relate to the three paragraphs with **two questions on each paragraph**. Each question has **only one** correct answer among the four given options (A), (B), (C) and (D).

Paragraph for Question Nos. 11 to 12

Read the following write up carefully and answer the following questions:

$$\int_a^b f(x)d(\alpha(x)) - \int_a^b \alpha(x).d(f(x)) = \alpha(b)f(b) - \alpha(a).f(a)$$

11. $\int_0^6 (x^2 + [x])d(|3 - x|)$ (where [.] denotes greatest integer function) is equal to

(A) 126 (B) 63
(C) 72 (D) 36

12. $\int_{-2}^{3} [|x|]d(|x|)$, (where [.] denote greatest integer function) is equal to

(A) 1 (B) 2
(C) 4 (D) none of these

Paragraph for Question Nos. 13 to 14

Read the following write up carefully and answer the following questions:

Let f(x, y) be defined $\forall$ x, y ϵ N such that

(i) $f(x, x+1) = \frac{1}{3}$, $\forall$ x ϵ N

(ii) f(x, y) = f(x, z) + f(z, y) – 2f(x, z) f(z, y), for all z such that x < z < y; x, y, z ϵ N

13. $\lim_{n\to\infty} f(1, n)$ is equal to

(A) 0 (B) $\frac{1}{2}$
(C) $\frac{2}{3}$ (D) $\frac{1}{3}$

14. f(2, 5) is equal to

(A) $\frac{13}{27}$ (B) $\frac{8}{27}$
(C) $\frac{1}{27}$ (D) $\frac{1}{3}$

Space for rough work

Paragraph for Question Nos. 15 to 16

Read the following write up carefully and answer the following questions:

Locus of the centre of a circle touching the circle $x^2 + y^2 - 4y - 2x = \frac{7}{4}$ internally and tangents on which from (1, 2) is making an angle 60° with each other is director circle (*C*) of a variable ellipse E. On this basis given answers to the following questions

15. The equation of the director circle (C) is

(A) $(x-2)^2+(y-1)^2=\sqrt{3}$ (B) $(x-1)^2+(y-2)^2=3$

(C) $(x+1)^2+(y-2)^2=\sqrt{3}$ (D) $(x-1)^2+(y+2)^2=3$

16. Locus of focus of variable ellipse (E) when major axis is parallel to x-axis, is

(A) $\left\{(x, y)/\sqrt{\frac{3}{2}} < |x-1| < \sqrt{3} \text{ and } y=2\right\}$ (B) $\left\{(x, y)/0 < |x| < \sqrt{3} \text{ and } y=2\right\}$

(C) $\left\{(x, y)/0 < |x+1| < \sqrt{3} \text{ and } y=2\right\}$ (D) $\left\{(x, y)/\sqrt{\frac{3}{2}} < |x-1| < \sqrt{3} \text{ and } y=2\right\}$

(Match List Type)

This section contains **4 multiple choice questions.** Each question has matching lists. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

17. Match the following List-I with List-II

	List – I	List – II
(P)	If $f(x)=\int_{\pi/3}^{x}\frac{\sin t}{t}dt$ for $\frac{\pi}{2}\le x\le\pi$, then $\frac{\pi}{2}f\left(\frac{\pi}{2}\right)-\int_{\pi/3}^{\pi/2}f(x)dx$	1. $\frac{2}{3}$
(Q)	If $x+y+z=4$ and $x^2+y^2+z^2=6$, then least value of x is	2. 2
(R)	If all the two digit numbers from 19 to 92 are written consecutively to form the number N = 19202122……9192, the largest power of 3 that divides N is	3. $\frac{1}{2}$
(S)	If $z\in C$ such that $\lvert z-1\rvert+\lvert z+3\rvert\le 8$, then minimum value of $\frac{1}{\sqrt{3}}\lvert z+1\rvert$ is equal to	4. 1

Codes:

	P	Q	R	S
(A)	3	1	4	2
(B)	1	2	3	4
(C)	4	3	2	1
(D)	3	1	2	4

Space for rough work

18. Match the following List-I with List-II

List – I		List – II	
(P)	If the expression $\sin^2(\theta) + \sin^2(120° + \theta) + \sin^2(120° - \theta)$ remains constant, $\forall\theta \in R$, then the value of the constant is	1.	$\frac{2}{3}$
(Q)	If the sum of the solutions of the equation $2e^{2x} - 5e^x + 4 = 0$ is ln k then k equals	2.	1
(R)	If $\frac{\tan\theta}{\tan\theta - \tan 3\theta} = \frac{1}{3}$, find the value of $\frac{\cot\theta}{\cot\theta - \cot 3\theta}$	3.	$\frac{3}{2}$
(S)	If $3\tan(\theta - 30°) = \tan(\theta + 120°)$ then the value of $\cos 2\theta$ equals	4.	2

Codes:

	P	Q	R	S
(A)	4	3	2	1
(B)	3	4	1	2
(C)	3	4	2	1
(D)	1	2	3	4

19. Match the following List-I with List-II

List – I		List – II	
(P)	The distance of the point of intersection of the planes $\vec{r}.(2\hat{i} + \hat{j}) = 0$, $\vec{r}.(4\hat{j} + 3\hat{k}) = 24$, and $\vec{r}.(12\hat{i} - 7\hat{k}) = 24$ from the origin	1.	256
(Q)	The value of λ such that the planes $x - 2y - z = 10$, $3x + 3y + \sqrt{\lambda}\, z = 8$ and $2x - y + 2z = k$ are not able to make faces of a tetrahedron	2.	36
(R)	The product of distances of any vertex from the remaining vertices of a nine-sided regular polygon inscribed in a circle or radius $\sqrt{2}$ units	3	81
(S)	If on the complex plane the numbers $72 + 56i\sqrt{3}$, $a + b\, i\sqrt{3}$ ($a, b \in R$, $b > 0$) and 0 represent vertices of an equilateral triangle then the value of $(a + b)^2$	4.	144

Codes:

	P	Q	R	S
(A)	2	3	1	4
(B)	1	2	3	4
(C)	4	3	2	1
(D)	2	3	4	1

Space for rough work

20. Match the following List-I with List-II

	List – I	List – II
(P)	If angles A, B of ΔABC are roots of equation $\sqrt{3}\tan^2\frac{\theta}{2}-\sqrt{3}\lambda\tan\frac{\theta}{2}+(\sqrt{3}-\lambda)=0$, then C can $\left(\frac{2}{\sqrt{3}}<\lambda<\sqrt{3}\right)$ be equal to	1. $\frac{\pi}{6}$
(Q)	If $\frac{\sin A}{A}>\frac{2}{\pi}$ where A is a angle of ΔABC then A cannot be	2. $\frac{\pi}{3}$
(R)	In a ΔABC if $\sin A\cos B+\cos A\sin B\sin C=1$, then angle C can be	3. $\frac{\pi}{2}$
(S)	If $16\cos^4 x+k^2-2k=8\cos^2 x-2$ then possible values of $\frac{\pi k}{6}$ is	4. $\frac{2\pi}{3}$

Codes:

	P	Q	R	S
(A)	1	2	3	4
(B)	2	4	3	1
(C)	2	4	1	3
(D)	4	3	2	1

Space for rough work